

CONTROL STRUCTURE

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PROGRAM:

```
MySQL 8.0 Command Line Client
mysql> CREATE DATABASE ControlDB;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> USE ControlDB;
Database changed
mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION calculate_grade(marks INT)
-> RETURNS VARCHAR(5)
-> DETERMINISTIC
-> BEGIN
->     DECLARE grade VARCHAR(5);
->
->     IF marks >= 90 THEN
->         SET grade = 'A';
->     ELSEIF marks >= 75 THEN
->         SET grade = 'B';
->     ELSEIF marks >= 60 THEN
->         SET grade = 'C';
->     ELSEIF marks >= 50 THEN
->         SET grade = 'D';
->     ELSE
->         SET grade = 'F';
->     END IF;
->     RETURN grade;
-> END //
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE FUNCTION get_department(dept_id INT)
-> RETURNS VARCHAR(30)
-> DETERMINISTIC
-> BEGIN
->     DECLARE dept_name VARCHAR(30);
->
->     CASE dept_id
->     WHEN 1 THEN SET dept_name = 'CSE';
->     WHEN 2 THEN SET dept_name = 'ECE';
->     WHEN 3 THEN SET dept_name = 'IT';
->     WHEN 4 THEN SET dept_name = 'MECH';
->     ELSE SET dept_name = 'Unknown';
->     END CASE;
->     RETURN dept_name;
-> END //
Query OK, 0 rows affected (0.01 sec)
```

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MySQL 8.0 Command Line Client
->     WHEN 4 THEN SET dept_name = 'MECH';
->     ELSE SET dept_name = 'Unknown';
->     END CASE;
->     RETURN dept_name;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE PROCEDURE display_numbers()
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->
->     WHILE i <= 10 DO
->         SELECT i AS Number;
->         SET i = i + 1;
->     END WHILE;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE PROCEDURE repeat_numbers()
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->
->     REPEAT
->     SELECT i AS Number;
->     SET i = i + 1;
->     UNTIL i > 5
->     END REPEAT;
-> END //
Query OK, 0 rows affected (0.01 sec)
```

```
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mysql>
mysql> CREATE PROCEDURE loop_numbers()
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->
->     number_loop: LOOP
->         SELECT i AS Number;
->
->         SET i = i + 1;
->
->         IF i > 5 THEN
->             LEAVE number_loop;
->         END IF;
->     END LOOP number_loop;
-> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
mysql> DELIMITER ;
mysql>
```

OUTPUT:

```
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mysql> SELECT calculate_grade(95) AS Grade;
+-----+
| Grade |
+-----+
| A      |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> SELECT get_department(1) AS Department;
+-----+
| Department |
+-----+
| CSE        |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> CALL display_numbers();
+-----+
| Number |
+-----+
| 1       |
+-----+
1 row in set (0.00 sec)

+-----+
| Number |
+-----+
| 2       |
+-----+
1 row in set (0.01 sec)

+-----+
| Number |
+-----+
| 3       |
+-----+
1 row in set (0.01 sec)

+-----+
| Number |
+-----+
| 4       |
+-----+
```



```
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1 row in set (0.02 sec)

+-----+
| Number |
+-----+
|      5 |
+-----+
1 row in set (0.03 sec)

+-----+
| Number |
+-----+
|      6 |
+-----+
1 row in set (0.03 sec)

+-----+
| Number |
+-----+
|      7 |
+-----+
1 row in set (0.04 sec)

+-----+
| Number |
+-----+
|      8 |
+-----+
1 row in set (0.04 sec)

+-----+
| Number |
+-----+
|      9 |
+-----+
1 row in set (0.05 sec)

+-----+
| Number |
+-----+
|     10 |
+-----+
1 row in set (0.06 sec)

Query OK, 0 rows affected (0.06 sec)
```

```
MySQL 8.0 Command Line Client

Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> CALL repeat_numbers();
+-----+
| Number |
+-----+
|      1 |
+-----+
1 row in set (0.00 sec)

+-----+
| Number |
+-----+
|      2 |
+-----+
1 row in set (0.01 sec)

+-----+
| Number |
+-----+
|      3 |
+-----+
1 row in set (0.01 sec)

+-----+
| Number |
+-----+
|      4 |
+-----+
1 row in set (0.02 sec)

+-----+
| Number |
+-----+
|      5 |
+-----+
1 row in set (0.03 sec)

Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CALL loop_numbers();
+-----+
```

```
MySQL 8.0 Command Line Client
+----+
| Number |
+----+
|      5 |
+----+
1 row in set (0.03 sec)

Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CALL loop_numbers();
+----+
| Number |
+----+
|      1 |
+----+
1 row in set (0.00 sec)

+----+
| Number |
+----+
|      2 |
+----+
1 row in set (0.00 sec)

+----+
| Number |
+----+
|      3 |
+----+
1 row in set (0.00 sec)

+----+
| Number |
+----+
|      4 |
+----+
1 row in set (0.01 sec)

+----+
| Number |
+----+
|      5 |
+----+
1 row in set (0.01 sec)
```

